

Supporting Evidence

Attachment Rescue · Track 1 (Ideas). All figures reproducible from public data with the linked code.

1. CENSUS OF THE FULL POPULATION (FILES LISTED BY NAME, NOTHING DOWNLOADED)

NCI-funded open-access papers, 2016–2022, with supplementary files and no data-repository accession	Count
Papers in population (Europe PMC, funder "NCI NIH HHS"; no GEO/SRA/dbGaP/EGA/ArrayExpress/PRIDE/BioProject/MetaboLights/PDB accession)	30,376
Papers whose full text was readable · attachments listed	28,949 · 89,820
Papers with at least one spreadsheet / CSV / tab-text attachment	6,130 (21.2%)
Spreadsheet / CSV / tab-text files (Excel alone 19,818; 15.3 GB)	21,186
Those papers licensed CC BY or CC0 (files may be re-deposited unchanged)	4,673 (76%)
Share of papers with a spreadsheet attachment, by year 2016 → 2022	17% · 20% · 22% · 23% · 22% · 22% · 21%

2. FOUR RANDOM SAMPLES OF 300 PAPERS, EVERY ATTACHMENT DOWNLOADED AND OPENED

Cancer type	Population	Papers with a dataset*	Share	CC BY / CC0	Dataset files
Glioblastoma	630	61 / 300	20.3%	46	121
Breast cancer	3,104	65 / 300	21.7%	60	155
Colorectal cancer	888	67 / 300	22.3%	58	157
Leukemia	1,272	63 / 300	21.0%	43	148
Pooled		256 / 1,200	21.3% (95% CI 19.0–23.7%)	207 (81%)	581

*Dataset = largest table ≥ 50 rows $\times$ 3 columns, or $\geq 1,000$ rows; PDF/Word tables count only as numeric grids; peer-review files, reporting summaries and figures excluded. Largest table: 152,683 rows. Hand check of 105 machine decisions: 60/60 "dataset" calls were real data tables; 26/30 rejections correct, every miss a table under 50 rows. About half of rescued tables are primary measurements, half derived results.

3. TWO LIVE EXAMPLES A REVIEWER CAN OPEN

- **121-patient glioblastoma cohort table** (case, sex, age, location, IDH/TERT status, MGMT, ATRX) — *Acta Neuropathol Commun* 2018, PMC6193287, file 40478_2018_613_MOESM2_ESM.xlsx. CC BY. Listed in no catalogue.
- **118,273-row drug-screening dataset** — *Nat Commun* 2019, PMC6599391, file 41467_2019_10912_MOESM4_ESM.xlsx. CC BY. Listed in no catalogue.

4. DEPOSIT-READY PACKAGES

- 8 datasets from CC BY papers, unmodified, each with DataCite-style metadata.json (title, creators, license, article DOI, PMC ID, funder, format, size, sheet names) and README. Not yet uploaded; see governance in Prompt 4.

5. CODE, DATA, SITE (OPEN, MIT)

- Repository: github.com/OJ-IRO/attachment-rescue — census and classification code, raw census output, per-cancer results, packages.
- Public page with searchable list of the 581 dataset files and plain-language descriptions: attachment-rescue.vercel.app

6. PRIOR WORK CITED

- Gobeill et al., *Bioinformatics Advances* 2025 — 36.2 M PMC supplementary files extracted and indexed (PMC12371329).
- Federer et al., *PLOS ONE* 2018 (doi 10.1371/journal.pone.0194768); Colavizza et al., *PLOS ONE* 2020 (10.1371/journal.pone.0230416); Federer, 2022 (PMC9401135).
- Ghosh et al., *JNCI* 2025 (PMC12782234); Boja et al., *JCO CC* 2026 (PMC13479720); Boyd et al., *JNCI* 2026 (10.1093/jnci/djag199).
- CBIIT INS-Data pipeline: github.com/CBIIT/INS-Data; INS disease-code mapper: github.com/CBIIT/ccdi-ins-icdo-mapping-llm.